1. Core Correlation: Point 12

> **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**

- **Analysis:** This is the most prominent feature of the question. Students cannot find the answer in a single calculation; they must execute a **multi-step procedure**:
  1. Convert kilometers (km) to centimeters (cm) (unit conversion).
  2. Apply the division formula (total length ÷ total steps).
  3. Arrive at the final average step length.

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2. High Correlation: Point 11

> **Switch between different representations in the process of finding solutions**

- **Analysis:** The problem involves **switching between different units of representation**. Students must convert between the macro unit of "kilometers" and the micro unit of "centimeters" requested by the problem. This transition between different measurement representations is key to solving the problem.

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3. High Correlation: Point 7

> **Manipulate numbers, graphical and statistical data and information...**

- **Analysis:** Students need to **manipulate numerical information** (22,500 and 9) and perform mathematical operations based on the real-world context (pedometer data). This involves division of large numbers and understanding proportional relationships.

4. Moderate Correlation: Point 5

> **Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms**

- **Analysis:** Students need to call upon a **familiar algorithm**—specifically the algorithm definition of "average" (total amount / number of parts). At the same time, they must utilize the formal rules of length unit conversion (1 km = 100,000 cm) to ensure the calculation result aligns with system definitions.

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5. Moderate Correlation: Point 13

> **Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy**

- **Analysis:** The problem explicitly requires the answer in "centimeters (cm)". Students must understand the **contextual need** (step length is typically measured in centimeters, not meters or kilometers), and use this to guide the conversion process. A lack of understanding of context-appropriate precision could lead to an illogical answer.

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6. Lower Correlation: Point 1

> **Perform a simple calculation**

- **Analysis:** Although division is involved, the presence of unit conversion and multi-step logic means it transcends the scope of a "single simple calculation," which is why it is ranked below multi-step procedures.

1. 核心关联项：第 12 条

> **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation** (使用多步程序得出数学解、结论或泛化)

- **分析：** 这是该题最显著的特征。学生无法通过一步计算得出答案，必须执行**多步程序**：
  1. 将公里（km）转换为厘米（cm）（单位换算）。
  2. 应用除法公式（总长度 ÷ 总步数）。
  3. 得出最终的平均步长。

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2. 高度关联项：第 11 条

> **Switch between different representations in the process of finding solutions** (在寻找解决方案的过程中切换不同的表示形式)

- **分析：** 题目涉及**单位表示形式的切换**。学生必须在“公里”这一宏观单位和题目要求的“厘米”这一微观单位之间进行转换。这种在不同度量表示之间的切换是解题的关键 (p. 1)。

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3. 高度关联项：第 7 条

> **Manipulate numbers, graphical and statistical data and information...** (处理数字、图形和统计数据、信息.....)

- **分析：** 学生需要**处理数字信息**（22,500 和 9），并根据现实背景（计步器数据）进行数学操作。这涉及对大数字的除法运算以及对比例关系的理解。

4. 中度关联项：第 5 条

> **Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms** (理解并利用基于定义、规则和正式系统的结构，包括采用熟悉的算法)

- **分析：** 学生需要调用**熟悉的算法**——即“平均值”的算法定义（总量 / 份数）。同时，还需要利用长度单位换算的正式规则（1 km = 100,000 cm）来确保计算结果符合系统定义。

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5. 中度关联项：第 13 条

> **Use an understanding of the context to guide or expedite the mathematical solving process... working to a context-appropriate level of accuracy** (利用对背景的理解来指导或加速数学解决过程.....在适合背景的准确度水平上工作)

- **分析：** 题目明确要求以“厘米（cm）”为单位。学生必须理解**背景需求**（步长通常以厘米衡量而非米或公里），并以此指导换算过程。如果不理解背景对精度的要求，可能会得出不符合常理的答案 (p. 1)。

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6. 较低关联项：第 1 条

> **Perform a simple calculation** (进行简单的计算)

- **分析：** 虽然涉及除法，但由于存在单位换算和多步逻辑，它超越了“单一简单计算”的范畴，因此相关性排在多步程序之后。